Investigating Measles Transmission Dynamics in Texas with Kalman Filters

Malaika Shah[1,2], Farinaz Forouzannia[1,2], Lauren Ancel Meyers[1,2]
[1]The Oden Institute for Computational Engineering and Sciences, The University of Texas at Austin, [2]The University of Texas at Austin College of Natural Sciences

The University of Texas at Austin
College of Natural Sciences

ODEN INSTITUTE
FOR COMPUTATIONAL ENGINEERING & SCIENCES

Introduction

Measles is among the most contagious infectious diseases, with roughly nine out of ten unprotected individuals becoming infected upon exposure[1]. Between January and August 2025, Texas experienced a significant measles outbreak, resulting in over 750 reported cases and two deaths[2]. With vaccine rates declining across the US, outbreaks are occurring with increasing frequency[3]. Understanding the dynamics of these outbreaks and the efficacy of intervention can guide more effective public health responses.

In this study, we sought to estimate the baseline transmission rate and the extent to which public health responses reduce transmission throughout the early-2025 outbreak in Texas. We use an Extended Kalman Filter (EKF) to fit a disease transmission model to reported measles cases and deaths[4]. This process allows us to simultaneously estimate both observable and hidden state variables. The underlying model is a compartmental SIRD (Susceptible, Infected, Recovered, Deceased) model that incorporates immunity from vaccination.

We fit our model to both rolling-average and weekly-aggregated measles time series incidence data (Table 1). Both approaches suggest a rapid decline in transmission beginning in early March of 2025.

Research Goal

Use EKF to fit an epidemiological model to Texas 2025 measles incidence data to:

1. **Evaluate the approach:** Characterize EKF performance with noisy data
2. **Characterize the Outbreak:** Estimate the changing transmission rate

Methods

Data:
- Incidence and fatality data from the Texas Department of State Health Services[2]
- Data aggregation:
  - 7-day Rolling Average of Infections Data: computed a seven-day centered average of daily cases
  - Weekly Infections Data: aggregated cases over each week
- Vaccination Coverage: informed by county exemption rates[5], set to 97.6%
- Vaccination Efficacy: set to 95%[6]

Measles Infection Dynamic Model:
- SIRD Model: Figure 1 shows hidden (blue) and observable (orange) states
- Observations: Infected and fatality counts are directly measured
- Parameters:
  - β: measles transmission rate – estimated
  - δ: recovery rate – set to 0.15 days^{-1} (7 days infectious period)[7]
  - γ: mortality rate – set to a constant 2/210 days^{-1} , reflecting the number of deaths across the 7 month outbreak period
- Vaccination Rate & Efficacy: Incorporated by adjusting the population size

Kalman Filter for Data Assimilation:
- Algorithm: Applied EKF for case prediction and parameter estimation

Table 1. Two modeling approaches. Assimilation frequency refers to how often the model updated its predictions based on newly available data.

Model	Data	Assimilation frequency	Model timestep
Rolling	7-day rolling avg	Update estimates daily	1 day
Weekly	Weekly cases	Update estimates weekly	1 day

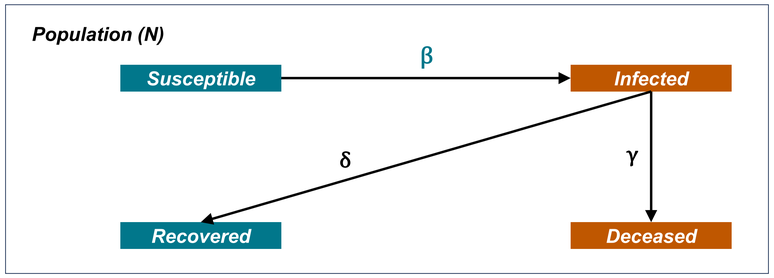


Figure 1. SIRD model diagram. Orange and blue represent observable states and hidden states, respectively. Approach assumes that all infections are reported.

State Differential Equations:

$$\frac{dS}{dt} = -\beta\frac{SI}{N};\ \frac{dI}{dt} = \beta\frac{SI}{N} - (\gamma+\delta)I;\ \frac{dR}{dt} = \gamma I;\ \frac{dD}{dt} = \delta I$$

Constraints:

$$N(\text{total population}) = S_t + I_t + R_t + D_t$$

$$\text{Where } N = \text{Texas Population}\left(1 - (\text{vaccine rate} * \text{vaccine efficacy})\right)^{8}$$

Figure 2. SIRD Model Formulation[4]. The model uses differential equations to track four dynamic states (boxes in Figure 1) and includes a constant vaccine rate and efficacy (refer to methods).

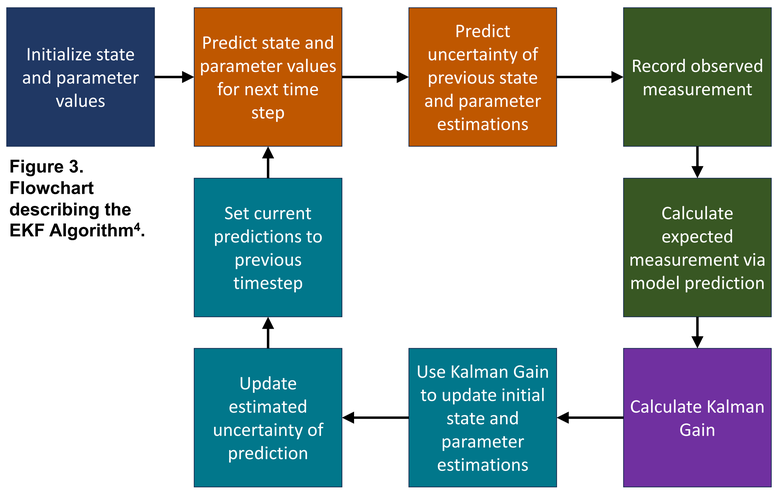


Figure 3. Flowchart describing the EKF Algorithm[4].

Figures and Results

Performance:
The rolling and weekly models achieved an R^2 of 0.99 and 0.86, respectively, measured by comparing estimated to observed incidence (Figure 4).

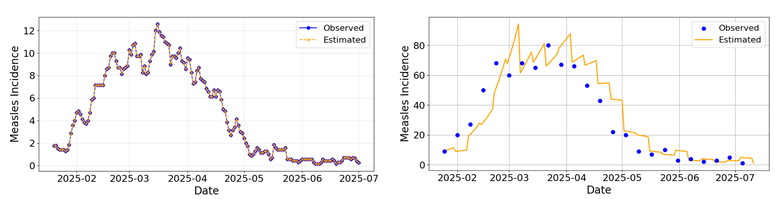


Figure 4. Comparison of observed and estimated incidence for rolling model (left) and weekly model (right)

Parameter Estimation:
Both models estimate an initial transmission rate of around 0.2-0.22, which corresponds to a basic reproduction number (R_0) of 1.3-1.5, and began declining exponentially at the beginning of March (Figure 5). The decline may reflect the impact of public health control measures including case isolation and local vaccination.

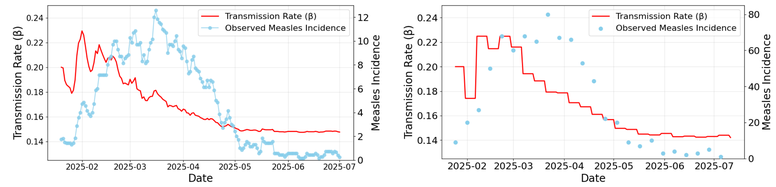


Figure 5. Estimated transmission rate from rolling model (left) and weekly model (right).

Discussion

This project applied an Extended Kalman Filter (EKF) to fit a compartmental SIRD epidemiological model to sparse measles data from Texas. The model produced incidence estimates that closely matched observed cases and inferred transmission rates that may reveal the timing and impact of public health interventions. It performed equally well when using data aggregated by a seven-day rolling window or by discrete weekly totals.

An important caveat is that the model assumes all measles cases are reported and does not account for undetected infections. Future work will incorporate unreported cases, structured interactions among school-aged children, and estimation of additional parameters like reporting and recovery rates.

Acknowledgments

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